Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 · Electrochemistry and corrosion · Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage)

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

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M1

Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage)

1 Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage)

Study focus: Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000182 · EV0000183

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module six on corrosion chemistry, and today we are going to study about some more factors affecting the rate of corrosion.

Key details

- The size of the anode and the cathode plays a major role in deciding the rate at which the anode undergoes corrosion.
- We remember from electrochemical theory that it's only the anode which undergoes corrosion and the cathode remains unaffected.
- Anode area being very small and a cathode area being very large, corrosion is very intensive and the anode will undergo corrosion at a very fast rate and as this ratio becomes smaller and smaller the corrosion rate is going to increase rapidly and this is what we studied under pitting corrosion.
- Is the reason for that.
- We know that when the anode is small, as in this case, and the cathode is large, the electrons are getting liberated at the anode and they are getting consumed at the cathode and since the cathode is so large there is rapid consumption of these electrons and the cathode and like and because of that demand for electrons is very high from the cathode.
- Therefore the anode has to undergo reaction at a very fast rate.
- That means it will get corroded at a very fast rate.
- If the tin coating is broken due to some reason, then we find that iron surface is getting exposed to the environment, and therefore, because iron is anode, it will be iron which will

undergo corrosion.
- And since we have a case of small anode and large case.
- Whereas if we have covered iron with zinc.
- The next factor is about hydrogen over voltage.
- Overvoltage is defined as the extra voltage which we need to provide over the theoretical voltage for a particular electrode reaction to take place.
- But we find that the reaction does not take place specially if there is evolution of gases.
- It is noticed that.
- In the case like say hydrogen gas or oxygen gas is getting evolved at the electrode due to the reaction, it is found that we need to provide this extra voltage, and that voltage is referred to as the overvoltage.
- And this overvoltage value depends on the.

Example or application from the lecture

- Let us study the ratio of anodic to cathodic area.
- For example, let us say we have an iron article which is covered with tin.
- So let us first understand what is overvoltage.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 05:00–09:59

Hydrogen over voltage values and also ox oxygen over voltage values.

Key details

- When we have just platinum, we have a value which is close to zero point one.
- But if you take the case of platinized platinum it's almost going to zero.
- That means the same hydrogen evolution on a platinum surface would not be would be more facilitated when it's platinized platinum why because the surface area as has become high so in the case of platinum we find that platinized platinum has lower hydrogen over voltage so what would be the result the hydrogen gas evolution which we will be much higher.
- This is the hydrogen voltage values.
- And then coming to platinum, it is even more easier.
- to for the hydrogen liberation and when it comes to platinized platinum it's even more the hydrogen over voltage is very less.
- This shows that hydrogen over voltage is dependent on the surface of the metal on which the hydrogen evolution is taking place.
- If a metal has low hydrogen overvoltage on its surface, then it is more susceptible to corrosion.
- Why would we say that?
- Because hydrogen liberation H plus electron is giving you half H2 is a cathodic reaction.
- Now, if in a particular during a particular corrosion, if cathodic reaction is hydrogen evolution type, then we will find that hydrogen gas gets liberated easily, and that means.
- The cathodic reaction is very fast.

- Now if the cathodic reaction is very fast, it is going to make the anodic reaction also fast because we know that anodic and cathodic reactions will take place simultaneously, and if the anode is undergoing reaction at a fast rate, it means that the high.
- The over overall corrosion is going to happen at a fast rate.
- And if the.
- Hydrogen overvoltage on the metal surface is high, that means the cathodic reaction has become slow and therefore the oxidation also becomes less and therefore the reac the corrosion of that metal will also become slower.
- We know that for any chemical reaction, as temperature increases, there is going to be an increase in the rate of reaction.
- And similarly, that is what we observe in the case of metals also or in the case of corrosion process.
- What are the various reasons?
- Apart from this, there are several other reasons which are going to promote corrosion due to high temperature.
- conductivity of the ionic conductance will increase with temperature.
- I have mentioned before, whether it's in batteries or in corrosion, it's not just the anode and cathode reaction which will decide the fate of the reaction.
- It's also the movement of ions in the electrolyte.
- the reaction will also be faster.
- We will study about this polarization soon and we will understand what is this polarization at the anode and cathode sites.

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Core explanation · 10:00–14:59

the because of high temperature polarization effect is less and therefore we find that corrosion will be faster.

Key details

- Also the s corrosion product solubility of the corrosion product generally increases at with high temperature.
- Solubility of any substance we know with increase in temperature it is going to increase.
- So here because this for the product is on the surface and if it is getting more dissolved into the solution into the medium we can understand that it's going to expose new surface every time and therefore we find that the rate of corrosion is also going to increase and apart from that another reason could be breakdown of the protective film.
- Now we know that many times an ice protective film gets formed over the surface but at a higher temperature it is possible that this the covering is going to the protective film is going to break down and therefore that could be another.
- Reason for height as the rate of reaction being higher or rate of corrosion being higher at a higher temperature.

- But there is an exception which we should know that oxygens like sorry, gases like oxygen carbon dioxide, if they.
- They dissolve, they are and are causing corrosion, then the rate of corrosion decreases with increase in temperature as the solubility of these gases in the medium decreases.
- So if the temperature is more than oxygen carbon dioxide, solubility is less and therefore we find that the rate of corrosion actually decreases.
- We know in general that when the pH is low, we expect that the rate of corrosion will be high, meaning that in acidic medium we find that the rate of corrosion is higher.
- Why is it so?
- Because when we have low pH.
- we find that the concentration of H plus ions is very high and we know that H plus ions are responsible for one of the cathodic reaction.
- If this H plus ion concentration is more, obviously we expect that this reaction will take place at a very fast rate.
- Now if the if this reaction happens fast, this being a cathodic reaction, it is also going to make the anodic oxidation faster and therefore we expect that the rate of corrosion will increase.
- Apart from that another reason in acidic medium is that corrosion product becomes more soluble in acetic medium then.
- Every time again a new surface is getting exposed and therefore we find that the rate of corrosion increases.
- But there is an exception here which we have to note that metals like aluminium actually undergo corrosion faster in highly alkaline solution.
- Whenever there is a drift in the electrode potential of the electrode, it is known as polarized.
- We say that the electrode got polarized.
- If it happens at anode we say it's anode has got polarized.
- If it happens at cathode we say cathode has got polarized.
- Electrode potential one of the important reasons could be a change in concentration.
- We know that when concentration changes, the electrode potential is also going to change.
- This here, look at the picture here, we have a metal surface, and at the metal, at the anode, we have anode and cathode getting formed on the surface.
- Metal is forming.
- Mn plus n electrons.

Example or application from the lecture

- Because we know that the corrosion product, for example, in case of aluminium A L2O3 is more soluble in alkaline medium.
- So even if L2O3 is getting formed, because it dissolves in the alkaline medium, we find that for aluminium in alkaline medium the rate of corrosion is going to be more let us look at another important factor that is called polarization of the anode and the cathode.
- First, let us understand what is polarization.

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EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 15:00–19:59

But many times the rate at which the anode is undergoing oxidation and the rate at which the metal ions are moving towards the cathode is different.

Key details

- Metal ions is very high.
- In that case, we find there is an accumulation of metalions around the anode, and when this happens, obviously the tendency for this reaction to go in the forward direction will be reduced, and that means the tendency of the metal to undergo.
- Oxidation is reduced, or in other words, corrosion will be less now.
- So the rate of corrosion is less, and the anode is said to be polarized because there is a drift in the electrode potential due to accumulation of ions.
- What we have to understand that whenever anodic polarization happens, it is going to reduce the rate of corrosion because the tendency of the metal to undergo oxidation is actually decreasing, the reason there is accumulation of ions around the anodic region.
- I reduction of oxygen or reduction of H plus ions, H plus ions taking up electrons, oxygen taking up electrons.
- When we find that there is not enough oxygen and H plus which are close to the cat hole.
- We find that there is a depletion of oxygen and H plus around the cathode, and therefore there is no reactant to pick up these electrons and that would result in cathodic polarization.
- Because there are less number of oxygen or H plus ions around the cathode, the rate of cathodic reaction also decreases.
- And therefore, we say that the cathode has got polarized and again this is resulting in slowing down of the corrosion product.
- And also it could be due to retarded removal of the product.
- When the anode and cathode get polarized we see that the rate of corrosion comes down.
- But in certain cases, like in batteries, etc., we don't want polarization.
- One the types of depolarizers are anodic depolarizers, depolarizers, sorry, not polarizers, and cathodic depolarizers.
- Similarly, cathodic depulverizers like cuplicons, which are going to take up the electrons themselves, and therefore this results in.
- Another way of in removing the polarization effect is to increase the temperature.
- Just now while studying temperature we understood that depolarization, we talked about the polarization effect if it is there, then.
- Then it is with increase in temperature, the polarization effect is decreased.
- We can understood understand this because with increase in temperature, there will be a decrease in polarization.
- Because now the ions are more temperature, their ions can move easily, the migration of ions can happen and therefore polarization will be removed.
- Also simple things like stirring the solution.
- Basically you are.

Example or application from the lecture

- Let us talk of cathodic polarization.

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EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 20:00–24:59

removing the ions from the metal ions from the anode or you are causing the material which has to undergo reduction, the species which has to undergo reduction to come closer to the capsule.

Key details

- Next, electrical conductivity of the corrosion medium.
- Just now we have discussed it several times that.
- More is the conductivity, faster will the ions move between the anode and the cathode region of the corrosion cell.
- Now when this happens, we know that the exchange of electrons at the electrode surfaces is going to happen very quickly, very fast, and therefore the corrosion rate is going to increase.
- Saline water what does it mean?
- It contains salt.
- And s we know are good ionic conductors and therefore when there is lot of ions in this in the water then we find that they will migrate very fast and therefore the exchange of electrons will also happen at the ion and cathode very fast and therefore the rate of corrosion will always is be high in saline water as compared to normal water or distilled water.
- The last factor we need to talk about is humidity.
- This is again understood that the very fact that electrochemical corrosion is happening, we know that we need a conducting medium.
- So what happens when the it is humid, the atmosphere is humid, then there is lot of moisture which acts as the conducting medium and it is a and we can get a nice galvanic cell over there and therefore this will facilitate corrosion of the metal.
- In arid areas we find therefore that corrosion will not happen to that great extent but whenever the atmosphere.
- Is moisture laden, then we find that corrosion will be fast because it is providing the conducting medium required for the migration of ions.
- And in the next class, now we are going to start talking about corrosion control.

Example or application from the lecture

- And there what we can one very good example would be that in saline water corrosion happens at a very fast rate.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

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A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

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2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00040	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

C Revision and retrieval practice